BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, FAISALABAD

INTERMEDIATE (12TH CLASS) EXAMINATION, 2024

COMPUTER SCIENCE

Past Paper 2024 • Paper Code: 8831

Retyped faithfully from the original board paper — same questions, no changes.

Computer Science — Objective

Time Allowed: 20 Minutes | **Maximum Marks:** 15 | **Paper Code:** 8831

Note: You have four choices for each objective type question as A, B, C and D. The choice which you think is correct, fill the relevant circle in front of that question number on computerized answer sheet. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero marks in that question. Attempt as many questions as given in objective type question paper and leave other circles blank.

Note: (MS-Access)

Q1. Storage and retrieval of data is related to:

(A) Data capturing
(B) Data manipulation
(C) Managing output results
(D) Data sorting

Q2. An attribute is also known as:

(A) Record
(B) Row
(C) Relation
(D) Field

Q3. Organizing the database in computer storage is done in:

(A) Logical design
(B) Physical design
(C) Analysis
(D) Implementation

Q4. A functional dependency is a relationship between two:

(A) Entities
(B) Relations
(C) Rows
(D) Attributes

Q5. A database consists of various components called:

(A) Objects
(B) Properties
(C) Tools
(D) Entities

Q6. The degree of relation refers to the number of:

(A) Rows
(B) Tables
(C) Fields
(D) Columns

> Note: (C - Language OR Visual Basic) — (C - Language)

Q7. The target code produced by the compiler is:

(A) Object code
(B) Source code
(C) Library code
(D) Linked code

Q8. The symbols that perform operations on data are called:

(A) Syntax
(B) Operands
(C) Operators
(D) Rules

Q9. An ampersand before the name of a variable in scanf ( ) function denotes:

(A) Address
(B) Actual value
(C) Format
(D) Data type

Q10. Format specifier %5d for the value 8135 will produce the output: [options (B)-(D) show leading blank spaces in the printed output]

(A) 8135
(B) [space]8135
(C) [space][space]8135
(D) [space][space][space]8135

Q11. The case block ends with:

(A) End select
(B) End case
(C) Case else
(D) Break

Q12. If X = 5 and Y = 3, the expression (X > Y) ? (X + Y) : (X - Y) ; will display output as:

(A) 8
(B) 2
(C) 0
(D) - 2

Q13. The do-while ( ) loop structure always ends with:

(A) Comma
(B) Colon
(C) Break
(D) Semi-colon

Q14. The scope of variable refers to its:

(A) Length
(B) Name
(C) Accessibility
(D) Data type

Q15. On successfully closing a file, the fclose ( ) returns:

(A) Null
(B) Zero
(C) One
(D) File pointer

> Note: OR (Visual Basic)

Q7. The step-by-step procedure to solve a problems is called:

(A) Algorithm
(B) Programming
(C) Control
(D) Flowchart

Q8. The bar that contains command needed to work with Visual Basic is called:

(A) Title Bar
(B) Status Bar
(C) Menu Bar
(D) Scroll Bar

Q9. The diamond symbol in a flowchart indicates:

(A) Condition
(B) Input
(C) Output
(D) Program

Q10. Control used to display text or to accept user input is called:

(A) Label
(B) Textbox
(C) Button
(D) Property

Q11. Forms and controls are collectively recognized as:

(A) Methods
(B) Properties
(C) Events
(D) Objects

Q12. Relational operators are used to:

(A) Compare
(B) Add
(C) Multiply
(D) Divide

Q13. Form property that displays text in the title bar of the form is:

(A) Title
(B) Header
(C) Border text
(D) Caption

Q14. In Visual Basic, which statement indicates the end of a decision structure?

(A) Next
(B) End
(C) End if
(D) Else

Q15. Which language is used for solving business problems?

(A) C++
(B) COBOL
(C) FORTRAN
(D) Assembly

Computer Science — Subjective

Time Allowed: 02:10 Hours | **Maximum Marks:** 60 | **Paper Code:** 8831

> Note: SECTION - I

Q2. (Marks 12) Write short answers to any SIX parts. (MS-Access)

i What is data dictionary?

ii Let you have a relation STUDENT in your database, such that STUDENT (Col1 , Col2 , Col3 , Col4 , Col5) create a view on first and last attribute of STUDENT.

iii What is data administrator?

iv What is recursive relationship? Give an example.

v Write any two disadvantages of "Centralized" data distribution strategy.

vi Define transitive dependency.

vii What is redundancy?

viii What is the main purpose of wildcards?

ix Differentiate between check boxes and radio buttons.

Q3. (Marks 12) Write short answers to any SIX parts. (C - Language)

i Define variable.

ii What is an array?

iii Find any two errors: int a = 10 , b = 40 , printf ("Sum = :" , a * b)

iv Find any two errors: Void main ( ) ; float i = 15.6; printf ("%d" , i) ;

v What is meant by format string?

vi Describe End-of-File (EOF).

vii Write the output of the following code: printf ("Escape sequence is a \" cool \ "feature of \'C-language\' ")

viii Find output: int a = 20; int b = 3; printf ("Result = %d" , a % b) ;

ix Write the statement in C-language: " Marks are between 40 and 60 "

> Note: OR (Visual Basic)

i Define property in VB.

ii What is the purpose of control naming conventions?

iii Describe code window.

iv Differentiate between runtime and break time.

v What are keyboard events in VB?

vi Write the purpose of the statement 'unload me'.

vii Define label control.

viii What is the purpose of a form?

ix How does two groups of option buttons work independently?

Q4. (Marks 12) Write short answers to any SIX parts. (C - Language)

i Define "switch" statement.

ii Write the syntax of "do-while" loop.

iii Predict the output of the following code segment: int x = 2 , y = 1 , z = 5; if (x > y && y > z) printf ("YES"); else printf ("NO");

iv Convert the following loop into do-while loop: int i = 1; while (i <= 15) { printf ("PUNJAB"); i ++; }

v List different types of functions in C.

vi Define function body.

vii Write the use of actual parameters.

viii How is a function activated?

ix Predict the output of the following code segment: int i = 5; do { printf ("PAKISTAN"); i = i - 1; } while (i > 0);

> Note: OR (Visual Basic)

i Define infinite loop in VB.

ii Define string data type in VB.

iii Write the syntax of Msgbox.

iv How select --- case helps in coding?

v What is default property of command button?

vi Define goto statement in VB.

vii Differentiate between pretest and post test loop.

viii What are legal characters of a identifier?

ix Write the syntax of For --- Next loop.

> Note: SECTION - II Each question carries 08 marks. (MS-Access) Attempt any ONE question from the following:

Q5. (Marks 08) Briefly describe four disadvantages of database systems.

Q6. (Marks 08) What is query? Discuss its uses and advantages.

> Note: (C - Language) Attempt any TWO questions either from C-Language OR Visual Basic

Q7. (Marks 02,02,02,02) What is basic structure of C-program? Explain its different parts briefly.

Q8. (Marks 08) Write a program in C-language that inputs marks of a student and displays his high grade according to the given criteria: Marks | >= 90 | 80 - 89 | 70 - 79 | 60 - 69 | 40 - 59 | < 40 Grade | A | B | C | D | E | F

Q9. (Marks 08) Write a program that displays first five numbers with their cubes using loop: Output 1 1 2 8 3 27 4 64 5 125

> Note: (Visual Basic)

Q7. (Marks 02,02,02,02) What are VB common controls? Explain any three common controls of VB.

Q8. (Marks 08) Write a program in Visual Basic that inputs marks of a student and displays his high grade according to the given criteria: Marks | >= 90 | 80 - 89 | 70 - 79 | 60 - 69 | 40 - 59 | < 40 Grade | A | B | C | D | E | F

Q9. (Marks 08) Write a program in VB that displays first five numbers with their cubes using loop: Output 1 1 2 8 3 27 4 64 5 125